Planning & Reinforcement Learning

Artificial Intelligence

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Indice

Parte I

Classical Planning

1 Introduction to Planning

1.1 Historical Context and Motivation

Planning has been a central topic in artificial intelligence since the inception of the field at the **Dartmouth Conference in 1956**, where the founding fathers of AI first gathered to define the scope and ambitions of this new discipline. Planning represents one of the fundamental capabilities required for intelligent behavior: the ability to reason about sequences of actions that achieve desired goals.

The motivations for studying automated planning are manifold:

- **Autonomous agents**: Enabling robots, software agents, and autonomous systems to make decisions and act independently in complex environments
- **Resource optimization**: Finding optimal or near-optimal strategies for resource allocation, scheduling, and logistics
- **Scientific applications**: Modeling and solving problems in domains such as space exploration, manufacturing, and healthcare
- **Theoretical foundations**: Understanding the computational complexity and formal properties of reasoning about action and change

1.2 The Planning Cycle

The complete planning process involves several interconnected phases that form a continuous cycle:

1. **Plan Generation**: Given a description of the current state, available actions, and desired goals, synthesize a sequence of actions (a plan) that transforms the initial state into a goal state. This phase involves:
    - Analyzing the initial state to understand what is currently true
    - Identifying the gap between the current state and the goal state
    - Searching through the space of possible action sequences
    - Evaluating candidate plans for correctness and optimality
    - Selecting the best plan according to specified criteria (e.g., shortest length, minimum cost, fastest execution)
2. **Plan Deployment**: Execute the generated plan in the actual environment, monitoring the execution to detect deviations from expected behavior. This phase includes:
    - Translating abstract plan actions into concrete executable commands
    - Monitoring sensors and feedback to verify that actions have the expected effects
    - Detecting discrepancies between predicted and actual states
    - Maintaining a record of executed actions for debugging and learning

3. **Replanning**: When execution monitoring detects that the current plan is no longer valid (due to unexpected events, action failures, or environmental changes), generate a new plan or repair the existing one. Replanning strategies include:
    - **Plan repair**: Modify the existing plan minimally to accommodate the new situation
    - **Replan from scratch**: Generate an entirely new plan from the current state
    - **Contingency planning**: Use pre-computed alternative plans for anticipated failures

This cycle reflects the reality that planning systems must operate in dynamic, partially unpredictable environments where initial plans may need adaptation. The cycle continues iteratively until the goal is achieved or deemed unreachable.

1.2.1 Challenges in Real-World Planning

Real-world deployment of planning systems faces several challenges:

- **Execution uncertainty**: Actions may fail or have unexpected outcomes
- **Incomplete information**: The planner may not have complete knowledge of the environment
- **Dynamic environments**: The world may change while the plan is being executed
- **Computational constraints**: Planning must often occur in real-time with limited computational resources
- **Plan quality vs. planning time**: Trade-off between finding optimal plans and responding quickly

These challenges motivate the development of robust planning algorithms that can handle uncertainty, adapt to changes, and operate efficiently under resource constraints.

2 Formalization of the Planning Problem

2.1 State Transition Systems

The mathematical foundation of planning rests on the concept of a **state transition system**, which provides a formal model of how actions transform states.

Definizione 2.1 (State Transition System)**.** A state transition system is a tuple $\Sigma = \langle S, A, \gamma \rangle$ where:

- S is a finite or countably infinite set of **states**
- A is a finite or countably infinite set of **actions**
- $\gamma : S \times A \to S$ is a **state transition function** that maps a state and an action to a resulting state

The transition function $\gamma(s, a) = s'$ specifies that executing action a in state s results in state s'. This function encodes the **dynamics** of the domain—how the world changes in response to actions.

2.1.1 Properties of State Transition Systems

State transition systems can be characterized by several important properties:

- **Determinism**: In a deterministic system, γ is a function—each state-action pair leads to exactly one successor state. In non-deterministic systems, multiple outcomes are possible.
- **Reachability**: A state s' is reachable from state s if there exists a sequence of actions that transforms s into s'. The reachable state space from an initial state is often much smaller than the total state space.
- **Reversibility**: An action is reversible if there exists another action that undoes its effects. Many real-world actions are irreversible (e.g., breaking an object).
- **State space structure**: The connectivity and topology of the state space significantly affect planning complexity. Highly connected spaces may have many solution paths, while sparse spaces may have few or no solutions.

2.2 Classical Planning: Fundamental Assumptions

Classical planning makes several simplifying assumptions that restrict the class of problems considered but enable efficient algorithmic solutions.

Classical Planning Framework

The classical planning framework relies on six key assumptions:

1. Finitely many states
2. Finitely many actions
3. Deterministic transition function
4. Full observability
5. Instantaneous actions
6. No exogenous events

1. **Finitely many states**: The state space S is finite, allowing exhaustive search techniques. This assumption ensures that the planning problem is decidable and that search algorithms will terminate.

Justification: While real-world domains may have infinite state spaces (e.g., continuous variables), finite approximations are often sufficient for practical purposes.

2. **Finitely many actions**: The action space A is finite, ensuring decidability. Each state has a finite branching factor in the state space graph.

   Justification: Even in complex domains, the number of distinct action types is typically manageable, though the number of ground actions (instantiated with specific objects) may be large.

3. **Deterministic transition function**: For every state s and action a, there is exactly one resulting state $\gamma(s, a)$. Actions have predictable, certain outcomes.

   Justification: Determinism simplifies reasoning about action effects. Non-deterministic planning requires more complex formalisms (e.g., Markov Decision Processes).

4. **Full observability**: The planner has complete knowledge of the current state at all times. There is no uncertainty about which state the system occupies.

   Justification: Full observability eliminates the need for belief state reasoning and sensing actions. Partial observability requires more sophisticated planning approaches.

5. **Instantaneous actions**: Actions have no duration; they occur instantaneously. Temporal reasoning is not required.

   Justification: Ignoring action durations simplifies the planning model. Temporal planning extends classical planning to handle durations and concurrent actions.

6. **No exogenous events**: Only the planning agent can change the world through deliberate actions. The environment remains static unless acted upon.

   Justification: Exogenous events (e.g., other agents, natural processes) introduce additional complexity. Classical planning assumes a static world between actions.

These assumptions, while restrictive, capture a significant and important class of planning problems and provide a foundation for understanding more complex, realistic planning scenarios.

2.2.1 Relaxing Classical Assumptions

Modern planning research has developed extensions that relax these assumptions:

- **Probabilistic planning**: Handles non-deterministic actions with probability distributions over outcomes
- **Conformant planning**: Plans under partial observability without sensing
- **Contingent planning**: Generates conditional plans that include sensing actions
- **Temporal planning**: Reasons about action durations and temporal constraints
- **Multi-agent planning**: Coordinates plans among multiple agents

2.3 The Planning Problem

Given a state transition system $\Sigma = \langle S, A, \gamma \rangle$, we can now formally define the planning problem.

Planning Problem

Definizione 2.2 (Planning Problem)**.** A planning problem is a tuple $P = \langle \Sigma, s_0, G \rangle$ where:

- $\Sigma = \langle S, A, \gamma \rangle$ is a state transition system
- $s_0 \in S$ is the **initial state**
- $G \subseteq S$ is a set of **goal states**

Alternatively, goals can be specified as a **goal formula** g, a logical expression that is satisfied by exactly those states in G. This allows more compact and expressive goal specifications.

2.3.1 Goal Specification

Goals can be specified in several ways:

- **Explicit goal states**: Enumerate the set G of acceptable final states. This is impractical for large state spaces.
- **Goal formula**: A logical formula g such that $G = \{s \in S \mid s \models g\}$. For example, in propositional logic: $g = \texttt{At}(\text{Robot}, \text{RoomB}) \wedge \texttt{Clean}(\text{RoomB})$
- **Goal conditions**: A set of propositions that must be true in the goal state. This is the most common approach in classical planning.
- **Utility functions**: In optimization settings, goals may be specified as utility functions to maximize or cost functions to minimize.

Solution Plan

Definizione 2.3 (Solution Plan)**.** A **solution** to a planning problem $P = \langle \Sigma, s_0, G \rangle$ is a sequence of actions $\pi = \langle a_1, a_2, \ldots, a_n \rangle$ such that:

$$s_n = \gamma(\gamma(\cdots \gamma(\gamma(s_0, a_1), a_2) \cdots, a_{n-1}), a_n) \in G$$

That is, executing the sequence of actions starting from the initial state s_0 results in a goal state.

We can also write this more compactly using the notation $\gamma^*(s, \pi)$ to denote the state reached by executing plan π from state s:

$$\gamma^*(s_0, \pi) \in G$$

2.3.2 Plan Quality

Not all solution plans are equally desirable. Common quality metrics include:

- **Plan length**: Number of actions in the plan. Shorter plans are often preferred.
- **Plan cost**: Sum of action costs. Each action a may have an associated cost $c(a)$.
- **Makespan**: Total execution time (relevant in temporal planning).
- **Resource consumption**: Amount of resources used during execution.

An **optimal plan** minimizes the chosen quality metric among all solution plans.

3 Representation of Planning Problems

3.1 The Challenge of Explicit Representation

While the state transition system formalism is mathematically elegant, it faces a critical practical challenge: **the curse of dimensionality**. Real-world planning domains can have exponentially large state spaces. For example:

- A domain with n boolean variables has 2^n possible states
- A domain with k objects and m binary relations has up to 2^{mk^2} states
- Enumerating all states and transitions explicitly is infeasible for even moderately-sized problems

Esempio 3.1 (Blocks World Complexity)**.** Consider a Blocks World domain with n blocks. The number of possible configurations grows super-exponentially with n. For $n = 10$ blocks, there are more than 10^{13} possible configurations. Explicitly representing the transition function would require storing information about trillions of state-action pairs.

This motivates the need for **compact, structured representations** that exploit regularities in the domain to represent large state spaces and transition functions implicitly.

3.2 Action Schemas: Structured Representation

The classical approach to compact representation uses **action schemas** (also called action templates or operators). An action schema is a parameterized description of a family of related actions.

Action Schema

Definizione 3.1 (Action Schema)**.** An action schema consists of:

- **Name and parameters**: A symbolic name and typed parameters, e.g., $\texttt{Move}(x, y)$
- **Preconditions**: A logical formula Pre specifying when the action can be executed
- **Effects**: A logical formula Eff specifying how the state changes when the action is executed

Actions are **ground instances** of action schemas, obtained by binding the parameters to specific objects in the domain. For example, the schema $\texttt{Move}(x, y)$ might generate ground actions $\texttt{Move}(\text{RoomA}, \text{RoomB})$, $\texttt{Move}(\text{RoomB}, \text{RoomC})$, etc.

3.2.1 Advantages of Action Schemas

Action schemas provide several benefits:

- **Compactness**: A single schema can represent exponentially many ground actions
- **Generality**: Schemas capture the general structure of actions independent of specific objects
- **Scalability**: Adding new objects to the domain automatically generates new applicable actions
- **Knowledge reuse**: Schemas learned in one domain can be transferred to similar domains

3.3 State Representation: Propositional Logic

In classical planning, states are typically represented using **propositional logic**:

- A state is a set of **atoms** (propositional variables) that are true in that state
- Atoms not in the set are assumed false (**closed-world assumption**)
- Action preconditions and effects are logical formulas over these atoms

This representation allows:

- Compact encoding of structured states
- Efficient reasoning about action applicability and effects
- Use of logical inference techniques for plan generation

3.3.1 State Update Semantics

STRIPS Assumption

When an action is executed, the state is updated according to the action's effects:

- **Add list**: Atoms that become true after the action
- **Delete list**: Atoms that become false after the action
- **Frame axioms**: Atoms not mentioned in the effects remain unchanged

The **STRIPS assumption** (named after the Stanford Research Institute Problem Solver) states that only atoms explicitly mentioned in the effects change; all other atoms persist unchanged. This simplifies reasoning about action effects.

3.4 Example: Blocks World

Consider the classic **Blocks World** domain, one of the most studied domains in planning research.

3.4.1 Domain Description

The Blocks World consists of:

- A set of blocks that can be stacked on top of each other
- A table with unlimited space
- A robot arm that can pick up and put down blocks

Blocks World State Representation

State predicates:
- $\texttt{On}(x, y)$: Block x is directly on top of block y
- $\texttt{OnTable}(x)$: Block x is on the table
- $\texttt{Clear}(x)$: Block x has no blocks on top of it
- $\texttt{Holding}(x)$: The robot arm is holding block x
- $\texttt{ArmEmpty}$: The robot arm is not holding anything

3.4.2 Action Schemas

PickUp(x): Pick up block x from the table

Parameters: x (block)
Precondition: $\texttt{Clear}(x) \land \texttt{OnTable}(x) \land \texttt{ArmEmpty}$
Effect: $\texttt{Holding}(x) \land \neg\texttt{OnTable}(x) \land \neg\texttt{Clear}(x) \land \neg\texttt{ArmEmpty}$

PutDown(x): Put down block x on the table

Parameters: x (block)
Precondition: $\texttt{Holding}(x)$
Effect: $\texttt{OnTable}(x) \land \texttt{Clear}(x) \land \texttt{ArmEmpty} \land \neg\texttt{Holding}(x)$

Stack(x, y): Stack block x on top of block y

Parameters: x, y (blocks)
Precondition: $\texttt{Holding}(x) \land \texttt{Clear}(y)$
Effect: $\texttt{On}(x, y) \land \texttt{Clear}(x) \land \texttt{ArmEmpty} \land \neg\texttt{Holding}(x) \land \neg\texttt{Clear}(y)$

Unstack(x, y): Remove block x from on top of block y

Parameters: x, y (blocks)
Precondition: $\texttt{On}(x, y) \land \texttt{Clear}(x) \land \texttt{ArmEmpty}$
Effect: $\texttt{Holding}(x) \land \texttt{Clear}(y) \land \neg\texttt{On}(x, y) \land \neg\texttt{Clear}(x) \land \neg\texttt{ArmEmpty}$

3.4.3 Example Problem Instance

Initial state: Blocks A, B, C are all on the table

$$s_0 = \{\texttt{OnTable}(A), \texttt{OnTable}(B), \texttt{OnTable}(C), \texttt{Clear}(A), \texttt{Clear}(B), \texttt{Clear}(C), \texttt{ArmEmpty}\}$$

Goal: Stack the blocks in the order A on B on C

$$G = \{\texttt{On}(A, B), \texttt{On}(B, C), \texttt{OnTable}(C)\}$$

Solution plan:

1. $\texttt{PickUp}(B)$
2. $\texttt{Stack}(B, C)$
3. $\texttt{PickUp}(A)$
4. $\texttt{Stack}(A, B)$

This plan has length 4 and achieves the goal from the initial state.

Parte II

Advanced Planning Representations

4 Reduction to Propositional Satisfiability

4.1 Motivation

We continue our discussion of the formalization and representation of planning problems. A fundamental technique in automated planning is to **reduce the planning problem to the satisfiability problem of propositional logic**. This reduction allows us to leverage powerful SAT solvers to find plans efficiently.

Reduction to SAT

The key idea is to encode:

- The initial state as a propositional formula
- The goal conditions as a propositional formula
- The action effects and preconditions as propositional formulas
- The constraint that actions must form a valid sequence

If the resulting formula is satisfiable, the satisfying assignment corresponds to a valid plan.

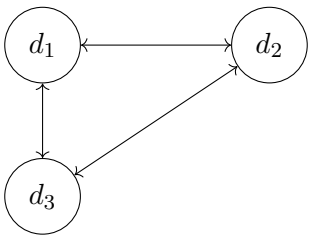
4.2 Example Domain: Dock Worker Robots (DWR)

To illustrate these concepts, we introduce the **Dock Worker Robots (DWR)** domain, a classic planning benchmark that models container logistics at a port.

4.2.1 Domain Description

The DWR world consists of:

- **Docks**: Three docks d_1, d_2, d_3 arranged in a triangular configuration
- **Connectivity**: Docks are connected as follows: $d_1 \leftrightarrow d_3$, $d_1 \leftrightarrow d_2$, $d_2 \leftrightarrow d_3$
- **Piles**: Storage locations at docks where containers can be stacked
- **Containers**: Objects that need to be moved between locations
- **Robots**: Mobile agents that can move containers between piles



4.2.2 Initial State Configuration

The initial state of our example problem is:

- **At dock** d_1: Pile p_1 contains two containers: c_1 (on top) and c_2 (below)
- **At dock** d_2:

  - Pile p_2 contains one container: c_3
  - Pile p_3 is empty
- **At dock** d_3: Pile p_4 is empty

4.3 Formal Representation

4.3.1 Objects and Types

The domain contains objects of different **sorts** (types). Each object is represented by a constant symbol with an associated type:

Docks:

$$\text{Docks} = \{d_1, d_2, d_3\}$$

where each d_i is a constant symbol of sort `Dock`.

Containers:

$$\text{Containers} = \{c_1, c_2, c_3\}$$

where each c_i is a constant symbol of sort `Container`.

Robots:

$$\text{Robots} = \{r_1, r_2\}$$

where each r_i is a constant symbol of sort `Robot`.

Piles:

$$\text{Piles} = \{p_1, p_2, p_3, p_4\}$$

where each p_i is a constant symbol of sort `Pile`.

The complete set of objects is:

$$\text{Objects} = \text{Docks} \cup \text{Piles} \cup \text{Containers} \cup \text{Robots}$$

4.3.2 Static Relations

Some relations in the domain are **static**—they do not change during plan execution. These can be specified once and remain constant throughout.

Adjacency relation: Specifies which docks are directly connected, allowing robots to travel between them.

$$\texttt{adjacent} = \{(d_1, d_3), (d_3, d_1), (d_2, d_3), (d_3, d_2), (d_1, d_2), (d_2, d_1)\}$$

This relation is symmetric: if dock d_i is adjacent to dock d_j, then d_j is adjacent to d_i.

Pile-Dock association: Each pile is permanently located at a specific dock.

$$\begin{gathered}\texttt{at}(p_1, d_1)\\ \texttt{at}(p_2, d_2)\\ \texttt{at}(p_3, d_2)\\ \texttt{at}(p_4, d_3)\end{gathered}$$

4.3.3 Rigid Relations

The adjacency relation is defined over $\texttt{Docks} \times \texttt{Docks}$. This is called a **rigid relation** (also known as a **static relation**)—it does not change during plan execution. Actions cannot modify the physical layout of the docks or their connectivity.

Rigid relations are important because:

- They reduce the state space by factoring out unchanging aspects
- They can be checked once at planning time rather than during execution
- They simplify action preconditions by providing fixed background knowledge

4.4 State Representation in DWR

4.4.1 What is a State?

A **state** is an assignment of values to propositional variables. More precisely, a state can be represented in two equivalent ways:

1. **First-order ground atoms**: Predicates where all arguments are constants (no variables)
2. **First-order ground terms**: Terms where all arguments are constants

Important restriction: We only allow **flat expressions**, meaning expressions with depth 1 (no nesting).

Esempio 4.1 (Forbidden Nesting)**.** Consider the predicate $\texttt{loc}(x, y)$ meaning "the location of x is y". We do **not** allow nested expressions such as:

$$\texttt{loc}(r_1, \texttt{loc}(r_2))$$

This would represent "the location of r_1 is the location of r_2", which violates the flatness constraint.

All atoms and terms must be flat—arguments of predicates and functions must be constants, not complex expressions.

4.4.2 State as Truth Assignment

A state assigns truth values to ground atoms. Consider describing the initial state s_0 of our DWR example:

$$\begin{gathered}\texttt{loc}(r_1, d_1) \\ \texttt{pos}(p_1, d_1) \\ \texttt{pos}(c_2, p_1)\end{gathered}$$

These atoms being listed in the state means they are **implicitly assigned true**:

$$\begin{aligned}\texttt{loc}(r_1, d_1) &\leftarrow \text{true} \\ \texttt{pos}(p_1, d_1) &\leftarrow \text{true} \\ \texttt{pos}(c_2, p_1) &\leftarrow \text{true}\end{aligned}$$

Atoms not mentioned in the state are assumed false (closed-world assumption).

4.5 Two Representation Formalisms

There are two common approaches to representing planning problems, differing in how they encode state information.

4.5.1 Classical (Propositional) Representation

In the **classical representation**, states are represented using only **Boolean assignments** to propositional atoms. Each possible fact about the world is represented as a Boolean variable.

Example atoms:

- $\texttt{loc}(r_1, d_1)$: Robot r_1 is at dock d_1 (true/false)
- $\texttt{pos}(c_2, p_1)$: Container c_2 is at pile p_1 (true/false)
- $\texttt{empty}(r_1)$: Robot r_1 is not carrying anything (true/false)

Characteristics:

- Simple and uniform representation
- Direct correspondence to propositional logic
- Easy to encode as SAT problems
- May require many atoms for complex domains

4.5.2 State Variable Representation

The **state variable representation** uses a slightly different syntax based on **function symbols**. Instead of Boolean atoms, we use variables that can take on different values from a finite domain.

Example:

$$\texttt{loc}(r_1) = d_1$$

This means "the location of robot r_1 is d_1". Here, $\texttt{loc}(r_1)$ is a **state variable** (a function that returns a value), and d_1 is its current value.

Comparison with classical representation:

Classical	State Variable
$\texttt{loc}(r_1, d_1) = \text{true}$ $\texttt{loc}(r_1, d_2) = \text{false}$ $\texttt{loc}(r_1, d_3) = \text{false}$	$\texttt{loc}(r_1) = d_1$

Advantages of state variables:

- More compact: one variable instead of multiple Boolean atoms
- Explicitly represents mutual exclusion (robot can only be at one location)
- Natural for domains with multi-valued attributes
- Reduces the number of variables in SAT encodings

Relationship: The two representations are **equivalent in expressive power**. Any problem representable in one formalism can be translated to the other. The choice depends on:

- The planning algorithm being used
- The structure of the domain

- Efficiency considerations for the specific problem

4.5.3 Example: Complete State Representation

Let us represent the complete initial state s_0 of our DWR example in both formalisms.

Classical representation (partial):

$$\begin{aligned}
&\texttt{loc}(r_1, d_1) = \text{true}\\
&\texttt{loc}(r_2, d_2) = \text{true}\\
&\texttt{pos}(c_1, p_1) = \text{true}\\
&\texttt{pos}(c_2, p_1) = \text{true}\\
&\texttt{pos}(c_3, p_2) = \text{true}\\
&\texttt{on}(c_1, c_2) = \text{true}\\
&\texttt{empty}(r_1) = \text{true}\\
&\texttt{empty}(r_2) = \text{true}\\
&\texttt{top}(c_1, p_1) = \text{true}\\
&\texttt{top}(c_3, p_2) = \text{true}
\end{aligned}$$

State variable representation:

$$\begin{aligned}
&\texttt{loc}(r_1) = d_1\\
&\texttt{loc}(r_2) = d_2\\
&\texttt{pos}(c_1) = p_1\\
&\texttt{pos}(c_2) = p_1\\
&\texttt{pos}(c_3) = p_2\\
&\texttt{on}(c_1) = c_2\\
&\texttt{loaded}(r_1) = \text{nil}\\
&\texttt{loaded}(r_2) = \text{nil}\\
&\texttt{top}(p_1) = c_1\\
&\texttt{top}(p_2) = c_3
\end{aligned}$$

Both representations capture the same information but organize it differently. The state variable representation is more compact and explicitly enforces constraints (e.g., a robot can only be at one location).

4.6 Key Insight: States as Boolean Assignments

The fundamental insight is that regardless of the representation formalism chosen:

A state is fundamentally a Boolean assignment

Whether we use propositional atoms or state variables, we are ultimately assigning truth values (or equivalently, values from finite domains) to a set of variables.

5 Action Schemas in DWR

5.1 Structure of Action Schemas

An **action schema** α is a template for generating ground actions. It consists of:

Action Schema Components

Definizione 5.1 (Action Schema)**.** An action schema α has the following components:

- **Name and parameters**: $\alpha : \texttt{name}(z_1, z_2, \ldots, z_k)$ where z_i are typed parameters
- **Preconditions**: $\text{Pre}(\alpha) = \{p_1, p_2, \ldots, p_n\}$ where each p_i is a literal
- **Effects**: $\text{Eff}(\alpha) = \{q_1, q_2, \ldots, q_m\}$ where each q_i is a literal
- **Cost**: $c(\alpha) \in \mathbb{N}_0$ (typically a non-negative integer)

Important constraint: The parameters z_i of α can appear in the precondition literals p_j and effect literals q_j. The arguments in these literals can be either:

- **Constants**: Specific objects from the domain
- **Parameters**: Variables that will be instantiated

No arbitrary free variables are allowed—all variables must be parameters of the action schema.

Ground Action

Definizione 5.2 (Ground Action)**.** A **ground action** a is obtained by substituting all parameters z_i in action schema α with specific constants from the domain. The action a is an **instance** of α.

5.2 Example: Take/Load Action

Consider the action of a robot taking a container from a pile. We define the action schema:

$$\alpha : \texttt{take}(r, c, c', p, d)$$

Parameters:

- r: robot (type `Robot`)
- c: container to take (type `Container`)
- c': container below c (type `Container` $\cup \{\texttt{nil}\}$)
- p: pile (type `Pile`)
- d: dock (type `Dock`)

5.2.1 Correcting Static Relations

Before defining the preconditions, we note that there are **two rigid (static) relations** in the DWR domain:

1. `adjacent` : `Docks` $\times$ `Docks` — specifies which docks are connected
2. `at` : `Piles` $\times$ `Docks` — specifies which pile is located at which dock

Both relations are rigid because the physical layout of the port does not change during plan execution.

5.2.2 Preconditions

For the `take` action to be executable, the following conditions must hold:

$$\text{Pre}(\texttt{take}(r, c, c', p, d)) = \left\{ \begin{array}{l} \texttt{loc}(r) = d \\ \texttt{at}(p, d) \\ \texttt{pos}(c) = p \\ \texttt{on}(c) = c' \\ \texttt{top}(p) = c \\ \texttt{cargo}(r) = \texttt{nil} \end{array} \right\}$$

Interpretation:

- The robot r is at dock d
- The pile p is located at dock d (static relation)
- Container c is positioned at pile p
- Container c is on top of container c' (or c' is `nil` if c is at the bottom)
- Container c is the top container of pile p
- The robot r is not carrying anything

5.2.3 Effects

When the `take` action is executed, the following changes occur:

$$\text{Eff}(\texttt{take}(r, c, c', p, d)) = \left\{ \begin{array}{l} \texttt{cargo}(r) = c \\ \texttt{top}(p) = c' \\ \texttt{pos}(c) = r \end{array} \right\}$$

Interpretation:

- The robot r is now carrying container c
- The top of pile p is now c' (the container that was below c)
- The position of container c is now the robot r

5.3 Action Applicability

Action Applicability

Definizione 5.3 (Action Applicability)**.** A ground action a is **applicable** (or **executable**) in state s if all preconditions of a are satisfied in s:

$$a \text{ applicable in } s \quad \Leftrightarrow \quad s \models \text{Pre}(a)$$

5.3.1 Examples

Consider two ground actions:

Example 1: $a_1 = \texttt{take}(r_1, c_1, c_2, p_1, d_1)$

In the initial state s_0 where:

- $\texttt{loc}(r_1) = d_1$
- $\texttt{at}(p_1, d_1)$

- $\texttt{pos}(c_1) = p_1$
- $\texttt{on}(c_1) = c_2$
- $\texttt{top}(p_1) = c_1$
- $\texttt{cargo}(r_1) = \texttt{nil}$

All preconditions are satisfied, so a_1 is **executable/applicable** in s_0.

Example 2: $a_2 = \texttt{take}(r_2, c_3, \texttt{nil}, p_4, d_3)$

This action requires:

- $\texttt{loc}(r_2) = d_3$
- $\texttt{at}(p_4, d_3)$
- $\texttt{pos}(c_3) = p_4$
- $\texttt{top}(p_4) = c_3$

However, in s_0, we have $\texttt{pos}(c_3) = p_2$ (not p_4) and $\texttt{loc}(r_2) = d_2$ (not d_3). Therefore, a_2 is **not executable/not applicable** in s_0.

6 State Transition Function

6.1 Formal Definition

The **state transition function** γ defines how actions transform states:

$$\gamma : \text{State} \times \text{Action} \to \text{State}$$

where $\gamma(s, a)$ computes the successor state obtained by executing ground action a in state s.

6.2 Computing the Successor State

Given a state s and a ground action a (which is a ground instance of an action schema α), the successor state $\gamma(s, a) = s'$ is defined if and only if:

1. $s \models p$ for all $p \in \text{Pre}(a)$ (all preconditions are satisfied)
2. s' is consistent (the resulting state contains no contradictions)

The successor state s' is computed as:

$$s' = \{q \mid q \in \text{Eff}(a)\} \cup \{q \mid s \models q, \neg q \notin \text{Eff}(a), q \notin \text{Eff}(a)\}$$

Interpretation: The new state consists of:

- All literals in the effects of a
- All literals from s that are not affected by a (neither the literal nor its negation appears in the effects)

6.3 Goal States

While the initial state is a specific assignment, goals are typically specified more flexibly.

Goal Specification

Definizione 6.1 (Goal Formula)**.** A **goal formula** G is a conjunction of literals specifying desired properties of goal states.

Definizione 6.2 (Goal States)**.** The set of **goal states** is:

$$S_g = \{s \mid s \models G\}$$

That is, all state assignments that satisfy the goal formula G.

6.4 Comparing Representations: Simplified DWR Example

Let us compare the classical and state variable representations using a simplified version of the DWR domain with three action schemas.

6.4.1 Classical Representation

In the classical representation, we use propositional atoms to represent facts:

Action 1: $\texttt{take}(r, c, l)$ — robot r takes container c at location l

$$\begin{aligned} \text{Pre:} &\quad \texttt{loc}(r, l) \wedge \texttt{pos}(c, l) \wedge \neg\texttt{loaded}(r) \\ \text{Eff:} &\quad \texttt{pos}(c, r) \wedge \neg\texttt{pos}(c, l) \wedge \texttt{loaded}(r) \end{aligned}$$

Action 2: $\texttt{move}(r, l, m)$ — robot r moves from location l to location m

$$\begin{aligned} \text{Pre:} &\quad \texttt{loc}(r, l) \\ \text{Eff:} &\quad \texttt{loc}(r, m) \wedge \neg\texttt{loc}(r, l) \end{aligned}$$

Action 3: $\texttt{put}(r, c, l)$ — robot r puts down container c at location l

$$\begin{aligned} \text{Pre:} &\quad \texttt{loc}(r, l) \wedge \texttt{pos}(c, r) \\ \text{Eff:} &\quad \texttt{pos}(c, l) \wedge \neg\texttt{pos}(c, r) \wedge \neg\texttt{loaded}(r) \end{aligned}$$

6.4.2 State Variable Representation

In the state variable representation, we use functional notation with assignments:

Action 1: $\texttt{take}(r, c, l)$

$$\begin{aligned} \text{Pre:} &\quad \texttt{loc}(r) = l \wedge \texttt{pos}(c) = l \wedge \neg\texttt{loaded}(r) \\ \text{Eff:} &\quad \texttt{pos}(c) \leftarrow r \wedge \texttt{loaded}(r) \end{aligned}$$

Action 2: $\texttt{move}(r, l, m)$

$$\begin{aligned} \text{Pre:} &\quad \texttt{loc}(r) = l \\ \text{Eff:} &\quad \texttt{loc}(r) \leftarrow m \end{aligned}$$

Action 3: $\texttt{put}(r, c, l)$

$$\begin{aligned} &\text{Pre:} && \texttt{loc}(r) = l \wedge \texttt{pos}(c) = r \\ &\text{Eff:} && \texttt{pos}(c) \leftarrow l \wedge \neg\texttt{loaded}(r) \end{aligned}$$

6.4.3 Key Differences

- **Classical**: Requires explicit deletion of old values (e.g., $\neg\texttt{loc}(r, l)$ when moving)
- **State Variable**: Implicit deletion through reassignment (e.g., $\texttt{loc}(r) \leftarrow m$ automatically removes the old location)
- **Compactness**: State variables are more concise for multi-valued attributes

7 Planning as Satisfiability (SAT)

7.1 Recap: Core Concepts

Before introducing the SAT encoding, let us recap the fundamental concepts:

- **State**: A Boolean assignment to propositional (ground) literals
- **Action**: Characterized by:
  - **Preconditions**: A set (or conjunction) of literals
  - **Effects**: A set (or conjunction) of literals

7.2 Temporal Instantiation of Variables

To encode planning problems as SAT, we introduce a temporal dimension to our propositional variables.

Temporal Instantiation

Definizione 7.1 (Temporal Instantiation)**.** Let X be the set of propositional variables describing the domain. The **temporal instantiation** of X at time t is:

$$X@t = \{x@t \mid x \in X\} \quad \text{for } t \geq 0$$

where $x@t$ is a propositional variable representing the truth value of x at time step t.

Intuition: Since a plan is a sequence of actions $(a_0, a_1, a_2, \ldots, a_{n-1})$ of length n, we need to track the truth value of each proposition at each time step from 0 to n.

7.3 The Bounded Planning Problem

Bounded Planning Problem

Definizione 7.2 (Bounded Planning Problem)**.** Given a planning problem $P = \langle \Sigma, I, G \rangle$, the **bounded planning problem** asks:
Does there exist a plan solving P with length exactly n?

This is a decision problem parameterized by the plan length n.

7.4 SAT Encoding

The bounded planning problem can be encoded as a propositional satisfiability problem.

Planning as SAT

Teorema 7.1 (Planning as SAT)**.** *For a planning problem P and bound n, there exists a propositional formula Φ_n such that:*

$$\Phi_n \text{ is satisfiable} \quad \Leftrightarrow \quad \text{there exists a plan of length } n \text{ for } P$$

The formula Φ_n encodes:

1. **Initial state**: $I@0$ (the initial state holds at time 0)
2. **Goal state**: $G@n$ (the goal holds at time n)
3. **Action preconditions and effects**: For each time step $t \in \{0, 1, \ldots, n-1\}$, exactly one action is executed and its effects are correctly applied
4. **Frame axioms**: Variables not affected by actions remain unchanged

7.5 Solving Planning via Iterative Deepening

Since we don't know the optimal plan length in advance, we use **iterative deepening**:

Iterative Deepening SAT Planning

Algorithm 1 Planning via SAT with Iterative Deepening

```
n ← 1
while true do
    Construct formula Φ_n
    if Φ_n is satisfiable then
        Extract plan from satisfying assignment
        return plan
    end if
    n ← n + 1
end while
```

Process:

1. Try $n = 1$: Check if Φ_1 is satisfiable
2. If not, try $n = 2$: Check if Φ_2 is satisfiable
3. If not, try $n = 3$: Check if Φ_3 is satisfiable
4. Continue until a satisfiable formula is found

Advantages:

- Finds shortest plans (optimal in number of steps)
- Leverages highly optimized SAT solvers
- Scales well for many planning domains

Challenges:

- Formula size grows with n
- May need to try many values of n before finding a solution
- Encoding frame axioms can be expensive

Parte III

SAT Solvers and Planning

8 The Boolean Satisfiability Problem

8.1 Definition and Scope

The **Boolean Satisfiability Problem (SAT)** is one of the most fundamental problems in computer science and forms the foundation for many automated reasoning techniques, including planning.

SAT Problem

Definizione 8.1 (SAT Problem)**.** The SAT problem asks: Given a propositional formula, does there exist an interpretation (truth assignment) that makes the formula true? Formally, given a set of clauses S in propositional logic, determine whether S is satisfiable.

8.2 Propositional Logic Basics

To understand SAT, we must first establish the basic elements of propositional logic:

- **Propositional variables**: Let X be a finite set of propositional variables, where each variable can assume one of two values: `true` (1) or `false` (0)
- **Atom (Atomic formula)**: A single propositional variable $x \in X$
- **Literal**: Either an atom x or its negation $\neg x$
- **Clause**: A disjunction of literals: $L_1 \vee L_2 \vee \cdots \vee L_n$ where each L_i is a literal

8.3 Interpretations and Truth Assignments

Interpretation

Definizione 8.2 (Interpretation)**.** An **interpretation** (or **truth assignment**) is a function:

$$v : X \to \{0, 1\}$$

that assigns a Boolean value to each propositional variable.

Number of interpretations: If $|X| = n$ (i.e., we have n propositional variables), then there are exactly 2^n possible interpretations.

8.3.1 Satisfiability of Clauses

Satisfaction

Definizione 8.3 (Clause Satisfaction)**.** A clause $C = L_1 \vee L_2 \vee \cdots \vee L_n$ is **satisfied** by an interpretation v if there exists at least one literal L_i such that $v(L_i) = \text{true}$ for some $i \in \{1, 2, \ldots, n\}$.
We write: $v \models C$
Definizione 8.4 (Formula Satisfaction)**.** A set of clauses $S = \{C_1, C_2, \ldots, C_m\}$ is **satisfiable** if there exists an interpretation v that satisfies all clauses in S:

$$v \models S \quad \Leftrightarrow \quad v \models C_i \text{ for all } i \in \{1, 2, \ldots, m\}$$

8.4 SAT Decision Procedure

The SAT problem can be viewed as a decision procedure with the following input-output specification:

$$S \text{ (set of clauses)} \xrightarrow{\text{SAT Solver}} \begin{cases} \textbf{YES (Satisfiable)} \\ \exists v : v \models S \\ \\ \textbf{NO (Unsatisfiable)} \\ \text{Proof: } S \vdash \square \end{cases}$$

Output cases:

1. **YES (Satisfiable)**: There exists an interpretation v such that $v \models S$. The SAT solver typically returns this satisfying assignment.
2. **NO (Unsatisfiable)**: No interpretation satisfies S. The solver may provide a proof of unsatisfiability, showing that S derives a contradiction (denoted $S \vdash \square$, where $\square$ represents the empty clause).

8.5 The Empty Clause

Empty Clause

Definizione 8.5 (Empty Clause)**.** The **empty clause**, denoted $\square$ (or $\square$), is a clause with no literals. It is **false in all interpretations** and therefore represents a contradiction.

Significance: If a SAT solver derives the empty clause from the input formula S, this constitutes a proof that S is unsatisfiable, since deriving a contradiction means no interpretation can satisfy all clauses.

9 Planning Problem Formalization

We formalized the planning problem with:

$$P = \langle \Sigma, s_0, S_g \rangle$$

where $\Sigma = \langle S, A, \gamma \rangle$.

- s_0 is the initial state

- S_g is the set of goal states: $S_g = \{s \mid s \models G\}$
- G is the goal formula (a conjunction of atoms)

9.1 States as Interpretations

States correspond to interpretations in propositional logic:

- $\texttt{at}(r_1, l_1)$: ground positive literal
- $\neg\texttt{loaded}(r_1)$: ground negative literal

Abstraction: From ground first-order literals to propositional literals. With finitely many ground atoms (literals) in the planning problem, we obtain finitely many variables $|X| = n$.

10 Bounded Planning Problem

Definizione 10.1 (Bounded Planning Problem)**.** Does there exist a plan π that solves planning problem P and has length k (i.e., $|\pi| = k$, where π is a sequence of k actions)?

10.1 Iterative Deepening on Plan Length

We use iterative deepening to find the shortest plan by increasing the bound k:

$$P \xrightarrow{\text{translation}} \Phi \xrightarrow{\text{SAT}} \begin{cases} \textbf{YES} : \text{interpretation } v \models \Phi \\ \text{encodes plan } \pi \text{ of length } k \\ \textbf{NO} : \text{no plan of length } k \text{ solves } P \end{cases}$$

The process iterates: $k := k + 1$ until a solution is found.

Key insight: The SAT encoding Φ includes:

- Initial state constraints at time 0
- Goal constraints at time k
- Action preconditions and effects for each time step
- Frame axioms ensuring persistence of unchanged literals

10.2 Temporal Encoding of Literals

A key aspect of the SAT encoding for planning is the **temporal instantiation** of literals. Since a plan of length k requires tracking state changes over time, each literal is indexed by time steps.

- **Plan length** k means we have $0 \leq i \leq k - 1$ time steps (transitions)
- Each literal f becomes $f@t$ where t ranges from 0 to k
- State i is the state after executing i actions (before the i-th action)

10.2.1 Example: Temporal Instantiation

Consider the literal $\texttt{at}(r_1, l_1)$ ("robot r_1 is at location l_1"):

$$\texttt{at}(r_1, l_1) \quad \xrightarrow{\text{temporal encoding}} \quad \texttt{at}(r_1, l_1, i)$$

Similarly, the negative literal $\neg\texttt{loaded}(r_1)$ becomes $\neg\texttt{loaded}(r_1, i)$, meaning "robot r_1 is not loaded at stage i".

State Transition

Definizione 10.2 (State Transition)**.** Step i (where $0 \leq i \leq k-1$) takes us from state i to state $i+1$ by executing one action.

10.3 Encoding the Initial State

The initial state s_0 is encoded as a conjunction of constraints at time $t = 0$:

$$\Phi_{s_0} = \bigwedge_{f \in s_0} f@0 \quad \wedge \quad \bigwedge_{f \notin s_0} \neg f@0$$

Where f represents ground literals.

10.3.1 From Literals to Propositional Variables

Each temporally indexed literal corresponds to a propositional variable:

- $\texttt{at}(r_1, l_1, 0)$ corresponds to variable $x_{\texttt{at}(r_1,l_1)}@0$
- $\neg\texttt{loaded}(r_1, 0)$ corresponds to $\neg x_{\texttt{loaded}(r_1)}@0$

10.3.2 Example: Complete Initial State Encoding

Consider an initial state containing $\texttt{at}(r_1, l_1)$ but not $\texttt{loaded}(r_1)$:

Literals in s_0 :	$\texttt{at}(r_1, l_1)$	$\wedge\neg\texttt{loaded}(r_1) \wedge \cdots$
Temporal encoding:	$\texttt{at}(r_1, l_1, 0)$	$\wedge\neg\texttt{loaded}(r_1, 0) \wedge \cdots$
SAT variables:	$x_{\texttt{at}(r_1,l_1)}@0$	$\wedge\neg x_{\texttt{loaded}(r_1)}@0 \wedge \cdots$

For literals not in the initial state (e.g., $\texttt{at}(r_1, l_2)$, $\texttt{at}(r_1, l_3)$), these are assigned false:

$$\neg\texttt{at}(r_1, l_2, 0), \quad \neg\texttt{at}(r_1, l_3, 0)$$

Alternative notation: The initial state encoding can also be written using set notation:

$$\Phi_{s_0} = \bigwedge_{x \in S_0^+} x_0 \wedge \bigwedge_{x \in S_0^-} \neg x_0$$

where S_0^+ is the set of positive literals in s_0 and S_0^- is the set of negative literals in s_0.

10.4 Encoding the Goal States

The goal conditions must be satisfied at the final time step k.

Goal Encoding

Definizione 10.3 (Goal Encoding)**.** The goal states are encoded at time k as:

$$\Phi_G = \bigwedge_{x \in G^+} x_k \wedge \bigwedge_{x \in G^-} \neg x_k$$

where G^+ is the set of positive literals in the goal G and G^- is the set of negative literals in G.

Interpretation: This formula ensures that all positive goal conditions are true and all negative goal conditions are false in the final state after executing k actions.

10.5 Encoding Actions

The planning problem is defined as $\Sigma = \langle S, A, \gamma \rangle$, where A is the set of all ground instances of the action schemas.

10.5.1 Action Preconditions

For each action $a \in A$ at each time step i (where $0 \leq i \leq k-1$), we encode the constraint that if the action is executed, its preconditions must hold:

Action Precondition Encoding

Definizione 10.4 (Action Precondition Encoding)**.**

$$\forall a \in A, \forall i \in \{0, \ldots, k-1\} : \quad a_i \rightarrow \bigwedge_{p \in \text{prec}(a)} p_i$$

where $\text{prec}(a)$ denotes the set of preconditions of action a.

Interpretation: This formula states that if action a is executed at time i, then all of its preconditions must be satisfied at time i. The implication can be rewritten in CNF as:

$$\neg a_i \vee \bigwedge_{p \in \text{prec}(a)} p_i$$

10.5.2 Mutual Exclusion of Actions

At each time step, at most one action can be executed. This constraint is encoded as follows:

Action Mutual Exclusion

Definizione 10.5 (Action Mutual Exclusion)**.** For all pairs of distinct actions $(a, b) \in A \times A$ where $a \neq b$, and for all time steps i where $0 \leq i \leq k-1$:

$$\neg a_i \vee \neg b_i$$

This formula ensures that actions a and b cannot both be executed at the same time step i.

Equivalent formulations:

- $a_i \rightarrow \neg b_i$: If action a is executed at time i, then action b is not executed
- $b_i \rightarrow \neg a_i$: If action b is executed at time i, then action a is not executed

10.6 Complete SAT Encoding

The complete SAT encoding Φ for a bounded planning problem with plan length k is the conjunction of the following sets of propositional formulas:

$$\Phi = \Phi_{s_0} \wedge \Phi_G \wedge \Phi_{\text{actions}} \wedge \Phi_{\text{mutex}} \wedge \Phi_{\text{frame}}$$

where:

- Φ_{s_0}: Initial state constraints
- Φ_G: Goal state constraints
- Φ_{actions}: Action precondition and effect constraints
- Φ_{mutex}: Mutual exclusion constraints between actions
- Φ_{frame}: Frame axioms (persistence of unchanged literals)

10.7 Plan Extraction from SAT Model

Once the SAT solver returns a satisfying assignment, we can extract the corresponding plan.

Plan Extraction

Definizione 10.6 (Plan Extraction)**.** Let v be a satisfying interpretation (model) returned by the SAT solver, represented as a set of literals. The plan π is extracted as follows:

$$\pi = \langle a_0, a_1, \ldots, a_{k-1} \rangle$$

where a_i is the unique action such that $v \models a_i$ (i.e., a_i is assigned `true` in the model v) for each time step $i \in \{0, 1, \ldots, k-1\}$.

Extraction procedure:

1. Filter the model v to obtain only the action literals: $\{a_i \mid a \in A, 0 \leq i \leq k-1, v \models a_i\}$
2. For each time step i, identify the action a such that a_i is true in v
3. Construct the sequence $\pi = \langle a_0, a_1, \ldots, a_{k-1} \rangle$

Nota. Due to the mutual exclusion constraints, at most one action is true at each time step. If no action is true at some time step i, this corresponds to a "no-op" (no operation) action.

Parte IV

SAT Encoding Examples

11 Detailed SAT Encoding Example

11.1 Simple Robot Navigation Domain

To illustrate the complete SAT encoding process, we present a simple example with a single robot and two locations.

Esempio 11.1 (Robot Navigation)**.** Consider a minimal planning domain with the following components:

Objects:

- One robot: r_1
- Two locations: l_1, l_2

Action schema: $\texttt{move}(r, l, l')$

Parameters:	r (robot), l (source location), l' (destination location)
Precondition:	$\texttt{at}(r, l)$
Effect:	$\texttt{at}(r, l') \land \neg\texttt{at}(r, l)$

11.1.1 Ground Actions

With one robot and two locations, we obtain two ground actions:

1. $a_1 = \texttt{move}(r_1, l_1, l_2)$: Move robot r_1 from location l_1 to location l_2
2. $a_2 = \texttt{move}(r_1, l_2, l_1)$: Move robot r_1 from location l_2 to location l_1

11.1.2 Problem Instance

Initial state (at time step $t = 0$):

$$s_0 = \{\texttt{at}(r_1, l_1)\}$$

The robot starts at location l_1. This is encoded temporally as:

$$\texttt{at}(r_1, l_1)@0$$

Goal state (at time step $t = 1$):

$$G = \{\texttt{at}(r_1, l_2)\}$$

The goal is for the robot to be at location l_2 after one action. This is encoded as:

$$\texttt{at}(r_1, l_2)@1$$

11.2 Complete SAT Encoding

For this problem with plan length $k = 1$, we construct the following propositional formulas:

11.2.1 Initial State Constraints

$$\Phi_{s_0} = \texttt{at}(r_1, l_1)@0 \land \neg\texttt{at}(r_1, l_2)@0$$

11.2.2 Goal State Constraints

$$\Phi_G = \texttt{at}(r_1, l_2)@1$$

11.2.3 Action Preconditions and Effects

For action $a_1 = \texttt{move}(r_1, l_1, l_2)$ at time step 0:

$$\texttt{move}(r_1, l_1, l_2)@0 \rightarrow \big(\texttt{at}(r_1, l_1)@0 \land \texttt{at}(r_1, l_2)@1 \land \neg\texttt{at}(r_1, l_1)@1\big)$$

For action $a_2 = \texttt{move}(r_1, l_2, l_1)$ at time step 0:

$$\texttt{move}(r_1, l_2, l_1)@0 \rightarrow \big(\texttt{at}(r_1, l_2)@0 \land \texttt{at}(r_1, l_1)@1 \land \neg\texttt{at}(r_1, l_2)@1\big)$$

11.2.4 Explanatory Frame Axioms

The **explanatory frame axioms** (also called **explanation axioms**) state that if a literal changes its truth value between time steps, then some action causing that change must have been executed.

If $\texttt{at}(r_1, l_1)$ becomes false:

$$\texttt{at}(r_1, l_1)@0 \land \neg\texttt{at}(r_1, l_1)@1 \rightarrow \texttt{move}(r_1, l_1, l_2)@0$$

If $\texttt{at}(r_1, l_2)$ becomes false:

$$\texttt{at}(r_1, l_2)@0 \land \neg\texttt{at}(r_1, l_2)@1 \rightarrow \texttt{move}(r_1, l_2, l_1)@0$$

If $\texttt{at}(r_1, l_1)$ becomes true:

$$\neg\texttt{at}(r_1, l_1)@0 \land \texttt{at}(r_1, l_1)@1 \rightarrow \texttt{move}(r_1, l_2, l_1)@0$$

If $\texttt{at}(r_1, l_2)$ becomes true:

$$\neg\texttt{at}(r_1, l_2)@0 \land \texttt{at}(r_1, l_2)@1 \rightarrow \texttt{move}(r_1, l_1, l_2)@0$$

11.3 Solution Extraction

Given the constraints, the SAT solver will find a satisfying assignment that includes:

$$\texttt{move}(r_1, l_1, l_2)@0 = \text{true}$$

The extracted plan is:

$$\pi = \langle \texttt{move}(r_1, l_1, l_2) \rangle$$

This plan has length 1 and achieves the goal of moving the robot from l_1 to l_2.

12 Conversion to Conjunctive Normal Form

12.1 CNF Definition and Structure

To use a SAT solver, we must convert all formulas to **Conjunctive Normal Form (CNF)**.

Conjunctive Normal Form (CNF)

Definizione 12.1 (Clause)**.** A **clause** is a disjunction of literals: $L_1 \vee L_2 \vee \cdots \vee L_k$ where each L_i is a literal (an atom or its negation).
Definizione 12.2 (Conjunctive Normal Form (CNF))**.** A formula is in **Conjunctive Normal Form** if it is a conjunction of clauses:

$$\text{CNF} = (L_1^1 \vee L_2^1 \vee \cdots \vee L_k^1) \wedge (L_1^2 \vee L_2^2 \vee \cdots \vee L_k^2) \wedge \cdots \wedge (L_1^m \vee L_2^m \vee \cdots \vee L_k^m)$$

A set of clauses is logically equivalent to a conjunction of those clauses, which is precisely the CNF representation required by SAT solvers.

12.2 Standard CNF Transformation

The standard transformation to CNF involves applying logical equivalences to eliminate implications and biconditionals, push negations inward, and distribute disjunctions over conjunctions.

12.2.1 Logical Equivalences

The following logical equivalences are used in the transformation:

Biconditional elimination:

$$F \leftrightarrow G \equiv (F \rightarrow G) \wedge (G \rightarrow F)$$

Implication elimination:

$$F \rightarrow G \equiv \neg F \vee G$$

De Morgan's laws:

$$\begin{aligned} \neg(F \vee G) &\equiv \neg F \wedge \neg G \\ \neg(F \wedge G) &\equiv \neg F \vee \neg G \end{aligned}$$

12.2.2 Negation Normal Form (NNF)

After applying the above equivalences, we obtain a formula in **Negation Normal Form (NNF)**, where negations appear only directly in front of atoms (not in front of complex formulas).

12.2.3 Distribution to Obtain CNF

To convert from NNF to CNF, we apply the **distributive law**:

$$F \vee (G \wedge H) \equiv (F \vee G) \wedge (F \vee H)$$

This distributes disjunctions over conjunctions, eventually producing a conjunction of disjunctions (CNF).

12.3 The Exponential Blowup Problem

Critical issue: The standard CNF transformation can cause a **combinatorial explosion** in formula size.

Exponential Blowup Problem

Osservazione (Exponential Blowup). In the worst case, the standard transformation into CNF produces a formula whose size is **exponential** in the size of the original formula: $O(2^n)$ where n is the size of the input formula.

This exponential blowup makes the standard transformation **impractical** for large planning problems, as the resulting CNF formula becomes too large to handle efficiently.

12.4 Equisatisfiable CNF (ECNF)

To avoid the exponential blowup, we use a different approach that produces an **Equisatisfiable CNF (ECNF)**.

Equisatisfiable CNF (ECNF)

Definizione 12.3 (Equisatisfiable CNF)**.** An **Equisatisfiable CNF (ECNF)** is a CNF formula that is not logically equivalent to the original formula, but is **equisatisfiable**—it has the same satisfiability status:

- The original formula is satisfiable if and only if the ECNF is satisfiable
- The ECNF may contain additional auxiliary variables not present in the original formula

12.4.1 Tseitin Encoding

The standard technique for producing ECNF is the **Tseitin encoding** (also called **Tseitin transformation**), named after Grigori Tseitin.

Key idea: Introduce auxiliary variables to represent subformulas, avoiding the need for distribution.

Tseitin Encoding Complexity

Teorema 12.1 (Tseitin Encoding Complexity)**.** *The Tseitin encoding produces an ECNF formula whose size is **linear** in the size of the original formula: $O(n)$ where n is the size of the input formula.*

Advantages of ECNF:

- **Linear size**: Avoids exponential blowup
- **Preserves satisfiability**: The ECNF is satisfiable if and only if the original formula is satisfiable
- **Efficient encoding**: Can be computed in polynomial time
- **Practical for planning**: Makes SAT-based planning feasible for large problems

Trade-off:

- The ECNF introduces additional variables (one for each subformula)
- The ECNF is not logically equivalent to the original formula (only equisatisfiable)

- However, the satisfying assignments to the original variables correspond to solutions of the original formula

12.5 Tseitin Encoding: Formal Construction

The Tseitin encoding is based on two key functions that work together to produce the ECNF.

12.5.1 The Representative Function

Representative Function

Definizione 12.4 (Representative Function)**.** Let PL denote the language of propositional formulas over a set of propositional variables X, and let V be a set of fresh (new) propositional variables. The **representative function** is defined as:

$$\text{Rep} : \text{PL} \to V \cup \{\bot, \top\}$$

The function Rep assigns to each formula a representative variable (or constant). It is defined inductively:

Base case (atomic formulas):

$$\begin{array}{lcll} \text{Rep}(x) & = & x & \text{for all } x \in X \\ \text{Rep}(\bot) & = & \bot & \\ \text{Rep}(\top) & = & \top & \end{array}$$

Inductive case (complex formulas):

$$\text{Rep}(F) = v_F$$

where $F \in \text{PL}$ is a non-atomic formula (i.e., F is neither a propositional variable nor $\bot$ nor $\top$), and $v_F \in V$ is a fresh propositional variable uniquely associated with F.

Intuition: Each complex subformula is represented by a new auxiliary variable. This avoids the need to duplicate subformulas during the transformation.

12.5.2 The Encoding Function

Encoding Function

Definizione 12.5 (Encoding Function)**.** The **encoding function** is defined as:

$$\text{En} : \text{PL} \to \text{PL}$$

$$\text{En}(F) = F \leftrightarrow \text{Rep}(F)$$

The encoding function creates a biconditional formula stating that formula F is logically equivalent to its representative.

Interpretation: $\text{En}(F)$ encodes the constraint that the auxiliary variable $\text{Rep}(F)$ must have the same truth value as the formula F itself.

12.5.3 Complete ECNF Construction

ECNF via Tseitin Encoding

Definizione 12.6 (ECNF via Tseitin Encoding)**.** Given an input formula F, let S_F denote the set of all subformulas of F (including F itself). The **Equisatisfiable CNF** of F is:

$$\text{ECNF}(F) = \text{Rep}(F) \wedge \bigwedge_{G \in S_F} \text{En}(G)$$

Components:

- $\text{Rep}(F)$: The representative variable for the entire formula (must be true)
- $\bigwedge_{G \in S_F} \text{En}(G)$: Conjunction of encoding constraints for all subformulas

Why "Equisatisfiable"?

The transformation from F to $\text{ECNF}(F)$ does **not** preserve logical equivalence, but it **does** preserve satisfiability:

Equisatisfiability Theorem

Teorema 12.2 (Equisatisfiability)**.** *Formula F is satisfiable if and only if ECNF(F) is satisfiable.*

Proof sketch. **Forward direction** (F satisfiable $\Rightarrow$ $\text{ECNF}(F)$ satisfiable): If F is satisfiable, we can extend the satisfying assignment to include the auxiliary variables by setting each $v_G = \text{value}(G)$ under the original assignment. This satisfies all encoding constraints.

Backward direction ($\text{ECNF}(F)$ satisfiable $\Rightarrow$ F satisfiable): If $\text{ECNF}(F)$ is satisfiable, the encoding constraints ensure that $\text{Rep}(F)$ has the same truth value as F. Since $\text{Rep}(F)$ must be true (it appears as a conjunct), F must also be true under the restriction of the assignment to the original variables. □

Key insight: The ECNF introduces auxiliary variables but maintains the same satisfiability status as the original formula, while keeping the formula size linear.

12.6 Example: Equisatisfiability vs. Logical Equivalence

To illustrate the difference between equisatisfiability and logical equivalence, consider a concrete example.

Esempio 12.1 (Tseitin Encoding of Disjunction)**.** Consider the formula:

$$F = A \vee (B \wedge C)$$

where A, B, and C are propositional variables.

Using the standard CNF transformation (distribution), we would obtain:

$$F_{\text{CNF}} = (A \vee B) \wedge (A \vee C)$$

This is **logically equivalent** to F, but for larger formulas, this approach leads to exponential blowup.

12.6.1 Tseitin Encoding Approach

Instead, we apply the Tseitin encoding by introducing an auxiliary variable v to represent the subformula $B \wedge C$:

Step 1: Define the encoding constraint:

$$v \leftrightarrow (B \wedge C)$$

Step 2: Replace the subformula in the original formula:

$$A \vee v$$

Step 3: Convert the biconditional to CNF. The biconditional $v \leftrightarrow (B \wedge C)$ expands to:

$$(v \rightarrow (B \wedge C)) \wedge ((B \wedge C) \rightarrow v)$$

Converting each implication:

$$v \rightarrow (B \wedge C) \equiv \neg v \vee (B \wedge C) \equiv (\neg v \vee B) \wedge (\neg v \vee C)$$
$$(B \wedge C) \rightarrow v \equiv \neg(B \wedge C) \vee v \equiv \neg B \vee \neg C \vee v$$

Final ECNF: The complete set of clauses is:

$$S = \{(A \vee v),\ (\neg v \vee B),\ (\neg v \vee C),\ (\neg B \vee \neg C \vee v)\}$$

Numbering the clauses:

$$\begin{array}{ll} (1) & A \vee v \\ (2) & \neg v \vee B \\ (3) & \neg v \vee C \\ (4) & \neg B \vee \neg C \vee v \end{array}$$

12.6.2 Demonstrating Equisatisfiability

Consider the interpretation:

$$I = \{A \mapsto \text{true},\ B \mapsto \text{false},\ C \mapsto \text{false}\}$$

Evaluating F under I:

$$F = A \vee (B \wedge C) = \text{true} \vee (\text{false} \wedge \text{false}) = \text{true} \vee \text{false} = \text{true}$$

Therefore, $I \models F$ (interpretation I satisfies F).

Evaluating S under I:

We need to extend I to assign a value to the auxiliary variable v. Let's first try without extending:

- Clause (1): $A \vee v = \text{true} \vee v = \text{true}$ (satisfied regardless of v)
- Clause (2): $\neg v \vee B = \neg v \vee \text{false} = \neg v$ (satisfied only if $v = \text{false}$)
- Clause (3): $\neg v \vee C = \neg v \vee \text{false} = \neg v$ (satisfied only if $v = \text{false}$)

- Clause (4): $\neg B \vee \neg C \vee v = \text{true} \vee \text{true} \vee v = \text{true}$ (satisfied regardless of v)

For I to satisfy S, we must set $v = \text{false}$ (to satisfy clauses (2) and (3)). With this extension:

$$I' = \{A \mapsto \text{true}, B \mapsto \text{false}, C \mapsto \text{false}, v \mapsto \text{false}\}$$

Now $I' \models S$ (all clauses are satisfied).

Equisatisfiability vs Logical Equivalence

Osservazione (Not Logically Equivalent). The original interpretation I (without the auxiliary variable) satisfies F but does **not** satisfy all clauses in S unless extended with the appropriate value for v. This demonstrates that F and S are **not logically equivalent**. However, F is satisfiable if and only if S is satisfiable (by appropriately choosing the value of v). Therefore, F and S are **equisatisfiable**.

Conclusion: The Tseitin encoding produces a formula that is equisatisfiable with the original but not logically equivalent. This is acceptable for SAT-based planning because we only care about whether a solution exists, not about preserving all logical properties.

12.7 Detailed Definition of the Encoding Function

We now provide the complete formal definition of the encoding function En for all cases.

Definizione 12.7 (Encoding Function - Complete Definition)**.** The encoding function $\text{En} : \text{PL} \to \text{PL}$ is defined inductively as follows:

Base cases (atomic formulas):

$$\begin{aligned} \text{En}(\bot) &= \top \\ \text{En}(\top) &= \top \\ \text{En}(x) &= \top \quad \text{for all } x \in X \end{aligned}$$

Inductive case for negation:

$$\begin{aligned} \text{En}(\neg F) &= \text{let } v = \text{Rep}(\neg F) \text{ in } v \leftrightarrow \neg\text{Rep}(F) \\ &= (v \to \neg\text{Rep}(F)) \wedge (\neg\text{Rep}(F) \to v) \\ &= (\neg v \vee \neg\text{Rep}(F)) \wedge (\text{Rep}(F) \vee v) \end{aligned}$$

Inductive case for binary connectives:

For all binary connectives $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$:

$$\text{En}(F_1 \circ F_2) = \text{let } v = \text{Rep}(F_1 \circ F_2) \text{ in } v \leftrightarrow (\text{Rep}(F_1) \circ \text{Rep}(F_2))$$

12.7.1 Example: Conjunction Encoding

Esempio 12.2 (Encoding Conjunction)**.** Consider the encoding of $F_1 \wedge F_2$:

$$\begin{aligned} \text{En}(F_1 \wedge F_2) &= \text{let } v = \text{Rep}(F_1 \wedge F_2) \text{ in } v \leftrightarrow (\text{Rep}(F_1) \wedge \text{Rep}(F_2)) \\ &= (v \to (\text{Rep}(F_1) \wedge \text{Rep}(F_2))) \wedge ((\text{Rep}(F_1) \wedge \text{Rep}(F_2)) \to v) \\ &= (\neg v \vee (\text{Rep}(F_1) \wedge \text{Rep}(F_2))) \wedge ((\text{Rep}(F_1) \wedge \text{Rep}(F_2)) \vee v) \end{aligned}$$

Distributing to obtain CNF:

$$(\neg v \vee \text{Rep}(F_1)) \wedge (\neg v \vee \text{Rep}(F_2)) \wedge (\text{Rep}(F_1) \vee v) \wedge (\text{Rep}(F_2) \vee v)$$

This produces **4 clauses** in CNF.

12.8 Complexity Analysis of Tseitin Encoding

We now analyze the size of the resulting ECNF formula to prove linear complexity.

Linear Size of ECNF

Teorema 12.3 (Linear Size of ECNF)**.** *Let F be a propositional formula with size $|F| = n$. The Tseitin encoding produces an ECNF formula:*

$$F' = \mathit{Rep}(F) \wedge \bigwedge_{G \in S_F} \mathit{En}(G)$$

where S_F is the set of all subformulas of F (including F itself). The size of F' is $|F'| = O(n)$.

12.8.1 Worst Case Analysis: Biconditional

The worst case for the encoding function occurs with the biconditional connective. Let us analyze:

Esempio 12.3 (Biconditional Encoding - Complete Derivation)**.** We show the complete step-by-step conversion of the biconditional encoding to CNF:

Step 1: Start with the encoding

$$\text{En}(F_1 \leftrightarrow F_2) = \text{let } v = \text{Rep}(F_1 \leftrightarrow F_2) \text{ in } v \leftrightarrow (\text{Rep}(F_1) \leftrightarrow \text{Rep}(F_2))$$

Step 2: Expand the inner biconditional

$$= v \leftrightarrow [(\text{Rep}(F_1) \to \text{Rep}(F_2)) \wedge (\text{Rep}(F_2) \to \text{Rep}(F_1))]$$

Step 3: Expand the outer biconditional

$$\begin{aligned} &= \big(v \to [(\text{Rep}(F_1) \to \text{Rep}(F_2)) \wedge (\text{Rep}(F_2) \to \text{Rep}(F_1))]\big) \\ &\quad \wedge \big([(\text{Rep}(F_1) \to \text{Rep}(F_2)) \wedge (\text{Rep}(F_2) \to \text{Rep}(F_1))] \to v\big) \end{aligned}$$

Step 4: Convert implications to disjunctions

$$\begin{aligned} &= \big(\neg v \vee [(\text{Rep}(F_1) \to \text{Rep}(F_2)) \wedge (\text{Rep}(F_2) \to \text{Rep}(F_1))]\big) \\ &\quad \wedge \big(\neg[(\text{Rep}(F_1) \to \text{Rep}(F_2)) \wedge (\text{Rep}(F_2) \to \text{Rep}(F_1))] \vee v\big) \end{aligned}$$

Step 5: Expand inner implications

$$\begin{aligned} &= \big(\neg v \vee [(\neg\text{Rep}(F_1) \vee \text{Rep}(F_2)) \wedge (\neg\text{Rep}(F_2) \vee \text{Rep}(F_1))]\big) \\ &\quad \wedge \big(\neg[(\neg\text{Rep}(F_1) \vee \text{Rep}(F_2)) \wedge (\neg\text{Rep}(F_2) \vee \text{Rep}(F_1))] \vee v\big) \end{aligned}$$

Step 6: Distribute $\neg v$ over the conjunction (first part)

$$= (\neg v \vee \neg\text{Rep}(F_1) \vee \text{Rep}(F_2)) \wedge (\neg v \vee \neg\text{Rep}(F_2) \vee \text{Rep}(F_1))$$

Step 7: Apply De Morgan's law to the second part

$$\wedge \big(\neg(\neg\text{Rep}(F_1) \vee \text{Rep}(F_2)) \vee \neg(\neg\text{Rep}(F_2) \vee \text{Rep}(F_1)) \vee v\big)$$

Step 8: Apply De Morgan's law again

$$\wedge \big((\text{Rep}(F_1) \wedge \neg\text{Rep}(F_2)) \vee (\text{Rep}(F_2) \wedge \neg\text{Rep}(F_1)) \vee v\big)$$

Step 9: Distribute disjunction over conjunction

$$\begin{aligned}&\wedge (\text{Rep}(F_1) \vee \text{Rep}(F_2) \vee v) \wedge (\text{Rep}(F_1) \vee \neg\text{Rep}(F_1) \vee v)\\ &\wedge (\neg\text{Rep}(F_2) \vee \text{Rep}(F_2) \vee v) \wedge (\neg\text{Rep}(F_2) \vee \neg\text{Rep}(F_1) \vee v)\end{aligned}$$

Step 10: Simplify (remove tautologies)

The clauses $(\text{Rep}(F_1) \vee \neg\text{Rep}(F_1) \vee v)$ and $(\neg\text{Rep}(F_2) \vee \text{Rep}(F_2) \vee v)$ are tautologies and can be removed.

Final CNF: The encoding produces the following clauses:

$$\begin{aligned}&(\neg v \vee \neg\text{Rep}(F_1) \vee \text{Rep}(F_2))\\ &(\neg v \vee \neg\text{Rep}(F_2) \vee \text{Rep}(F_1))\\ &(\text{Rep}(F_1) \vee \text{Rep}(F_2) \vee v)\\ &(\neg\text{Rep}(F_2) \vee \neg\text{Rep}(F_1) \vee v)\end{aligned}$$

This produces **4 clauses** after simplification (or up to 8 clauses before removing tautologies).

Proof of Linear Complexity. **Key observations**:

1. The number of subformulas $|S_F|$ is at most n (the size of F)
2. Each subformula $G \in S_F$ contributes at most a constant number of clauses (at most 8 for biconditional)
3. Therefore, the total number of clauses is at most $8n$
4. Each clause has at most 3 literals
5. Total size: $|F'| \leq 8n \times 3 = 24n = O(n)$

More precisely, we can bound the size as $|F'| = O(30n + 2)$, which is still linear in n. □

Conclusion: The Tseitin encoding achieves linear size $O(n)$ instead of the exponential size $O(2^n)$ of the standard CNF transformation, making it practical for large planning problems.